Two-Stage URL Cascade

One-shot identity-disjoint evaluation • 18,000 legitimate + 2,000 phishing URLs

STRICT 0.25% CALIBRATION BUDGET

System	Accuracy	Precision	Recall	F1	ROC-AUC	FPR	FNR
Single-stage GAN	94.51%	98.60%	45.75%	62.50%	97.52%	0.07%	54.25%
Two-stage cascade	96.81%	98.71%	69.05%	81.26%	96.43%	0.10%	30.95%

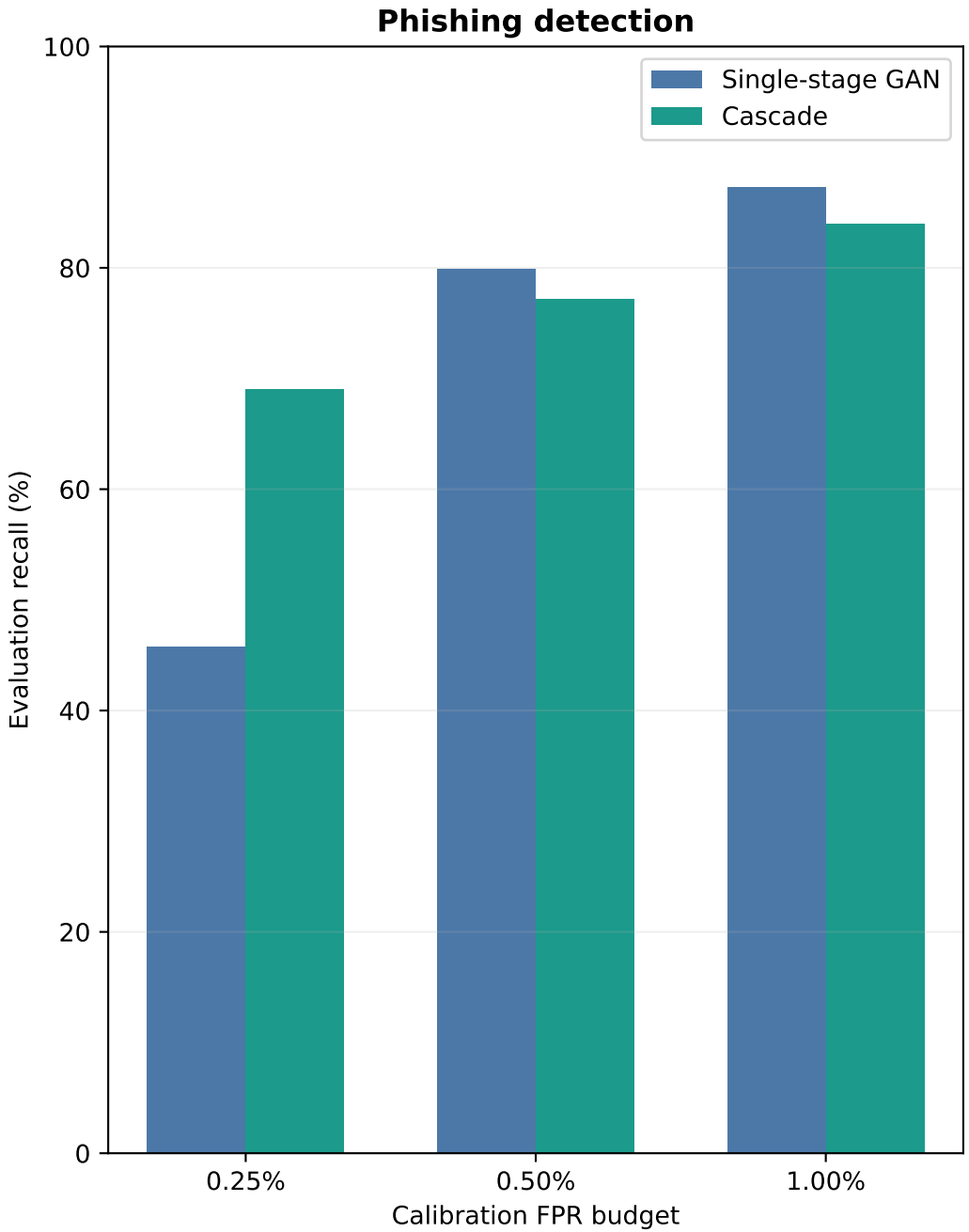
Recall delta: +23.30 points **FNR delta: -23.30 points**

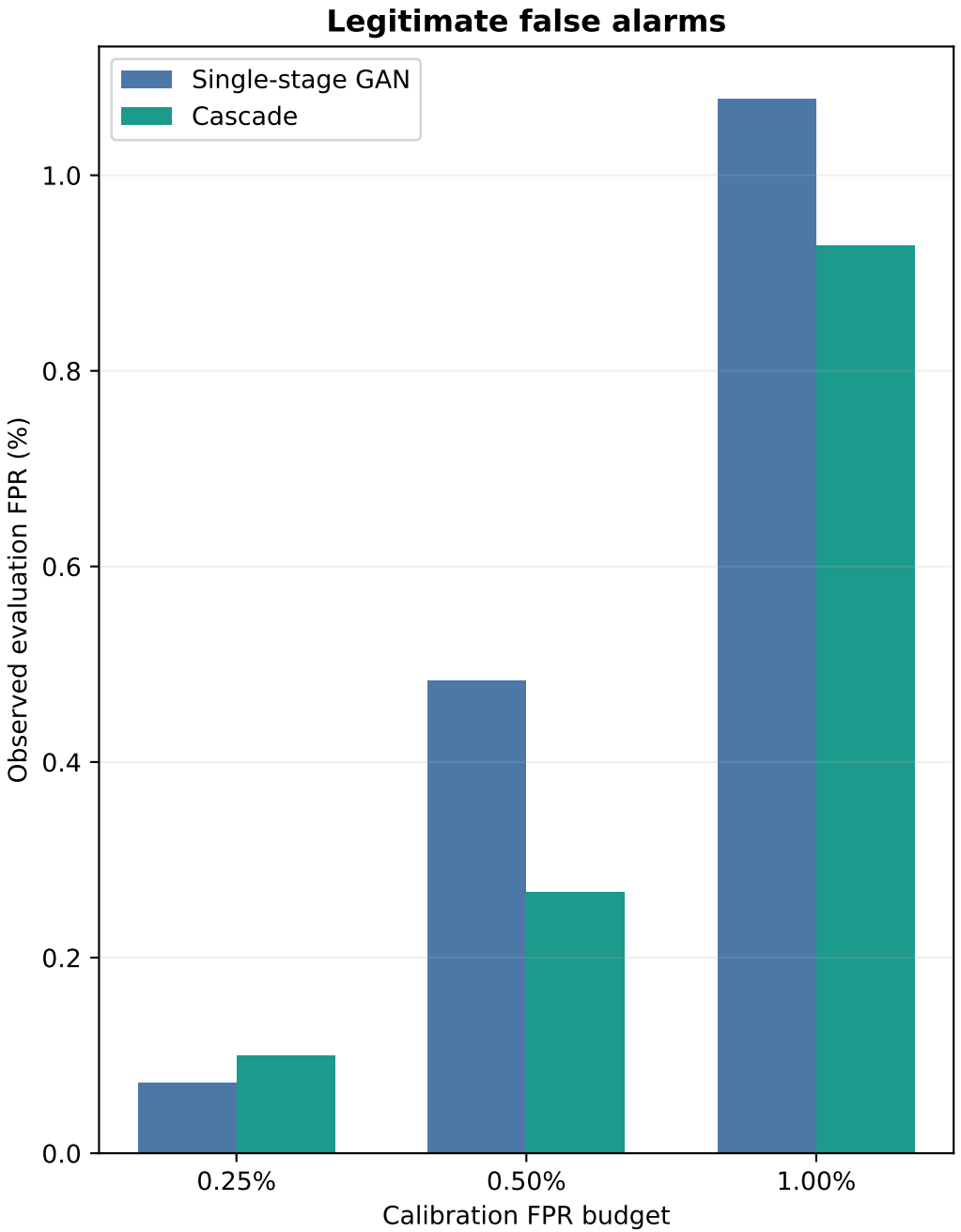
Observed counts — cascade: TP 1,381, FN 619, FP 18, TN 17,982

At the strict 0.25% calibration budget, the cascade changed recall by +23.30 points and FNR by -23.30 points; observed FPR changed by +0.028 points. This is evidence for the cascade only at this locked operating point, not proof that it eliminates false negatives.

This test was not used for training or threshold selection. It is not a later temporal snapshot and does not prove deployment read

Frozen operating points on the untouched cohort





What This Experiment Means

Architecture	Stage 1: updated HYB-MS +10% GAN screen. Stage 2: independent real-only lexical MLP verifier.
Strict result	At the strict 0.25% calibration budget, the cascade changed recall by +23.30 points and FNR by -23.30 points; observed FPR changed by +0.028 points. This is evidence for the cascade only at this locked operating point, not proof that it eliminates false negatives.
Routing	14.77% of URLs entered Stage 2 at the strict operating point.
AUC definition	Cascade ROC-AUC/PR-AUC use a continuous cascade decision score: immediate blocks rank first, routed rows use verifier scores, non-routed rows rank last.
Scope	Both stages remain URL-only. HTML, redirect chains, DNS age, TLS, reputation, and user context are absent.
Research conclusion	A second independent representation can reduce FNR under a very strict false-alarm budget, but its benefit must be judged per operating point and reconfirmed temporally.